

As such, the ratio of the maximum to minimum intensity will be $3^2/1^2 = 9$ times.

- 17** If a transparent glass block is placed in front of one of the slits of a Young's double slit experiment, which of the following observation of the fringe pattern will most likely occur?
- A** No fringe pattern will be observed on the screen.
 - B** The fringe separation will become larger.
 - C** The central maximum will be shifted from its original position.
 - D** The bright fringe will become brighter.

Ans: C

When the light passes through the glass block, it slows down and the wavelength becomes shorter. As such, more number of wavelengths would have elapsed if this wave were to travel to the original central maximum position as compared to the wave without the glass block. To ensure waves from both slits have no path difference when they meet at the central maximum, the position of the central maximum will have to be shifted closer to the slit with the glass block.

- 18** The diagram below shows two metal spheres **X** and **Y** which are placed close to each other